American Journal of Ophthalmology Case Reports 41 (2026) 102527

Contents lists available at ScienceDirect

American Journal of Ophthalmology Case Reports

journal homepage: www.ajocasereports.com/

Tubercular retinal vasculitis in widefield optical coherence tomography angiography

An-Lun Wu [a,b], Yukun Guo [a,c], Jinyi Hao [a,c], Erick Rivera-Grana [a], Steven T. Bailey [a], Yali Jia [a,c], Thomas S. Hwang [a,*]

[a] Casey Eye Institute, Oregon Health & Science University, Portland, OR, USA
[b] Department of Ophthalmology, Mackay Memorial Hospital, Hsinchu, Taiwan
[c] Department of Biomedical Engineering, Oregon Health & Science University, Portland, OR, USA

ARTICLE INFO

Keywords:
Tubercular retinal vasculitis
Retinal neovascularization
Nonperfusion area
OCT
OCT angiography

ABSTRACT

Purpose: To demonstrate the utility of widefield swept-source optical coherence tomography angiography (SS-OCTA) in identifying microvascular abnormalities in patients with presumed tubercular retinal vasculitis (TRV).
Observations: We present a case series of three patients with presumed TRV confirmed by positive QuantiFERON-TB Gold testing. All patients exhibited clinical signs of occlusive retinal vasculitis, including perivascular inflammation, peripheral capillary nonperfusion, and neovascular complications. Widefield SS-OCTA (DREAM OCT, Intalight) using a single-shot 26 × 21-mm scan revealed peripheral nonperfusion and vascular remodeling with features of retinal neovascularization, consistent with findings from fluorescein angiography (FA). Clinical improvement was observed following antitubercular therapy and anti-inflammatory treatment.
Conclusions and importance: Widefield OCTA provided detailed visualization of peripheral retinal changes associated with TRV, including nonperfusion area, collateral vessel formation, and neovascular tufts, highlighting its potential as a noninvasive complementary modality to FA in managing TRV.

1. Introduction

Tubercular retinal vasculitis (TRV) is an infectious cause of retinal vasculitis, primarily affecting retinal vessels with an occlusive nature.[1] Major vascular alterations are characterized by perivascular inflammation, retinal ischemia, and retinal neovascularization, which may result in severe vision loss if not promptly diagnosed and treated.[2] It presents a diagnostic challenge due to the absence of definitive microbiological confirmation in most cases, relying instead on clinical phenotype, immunological tests, and imaging features, particularly in nonendemic areas.[3] With the advent of optical coherence tomography angiography (OCTA), these vascular alterations can now be studied noninvasively.[4,5] Additionally, the recent development of widefield scans enables a more comprehensive evaluation of the retinal periphery, a crucial step in the assessment of TRV.

Herein, we report the application of this technique in three cases of retinal vasculitis with a presumptive tubercular etiology with a positive QuantiFERON-TB Gold test, analyzed using both single-shot 130-degree widefield swept-source OCTA (DREAM OCT, Intalight) volumetric scans of 26 × 21 mm with 1536 × 1240 sampling density and fluorescein angiography.

2. Cases presentations

2.1. Case 1

A 54-year-old male presented with a history of recurrent episodes of visual obscurations. He had been diagnosed with bilateral occlusive retinal vasculitis related to latent tuberculosis (TB) two years prior, treated at another hospital. Latent TB was confirmed by a positive QuantiFERON-TB Gold test, and the patient subsequently completed antitubercular therapy with rifapentine, isoniazid, and pyrazinamide. Prior to referral to our institution, he had been initiated on immunosuppressive therapy with azathioprine and prednisone for the management of retinal vasculitis, and had also undergone panretinal photocoagulation in both eyes.

On examination, the patient's best-corrected visual acuity was counting fingers at 2 feet in the right eye and 20/50 in the left eye. Slit-

* Corresponding author. Casey Eye Institute, Oregon Health & Science University, 515 SW Campus Dr, Portland, OR, 97239, USA.
E-mail address: hwangt@ohsu.edu (T.S. Hwang).

https://doi.org/10.1016/j.ajoc.2026.102527
Received 15 August 2025; Received in revised form 12 November 2025; Accepted 10 January 2026
Available online 30 January 2026
2451-9936/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

lamp biomicroscopy revealed no signs of anterior segment inflammation in either eye. Fundus examination revealed retinal neovascularization with old vitreous hemorrhage in the right eye and attenuated retinal vessels in the left (Fig. 1A). Ultra-widefield fluorescein angiography (FA) demonstrated widespread capillary nonperfusion with extensive neovascularization in the right eye, and mild macular leakage with multiple areas of capillary nonperfusion in the left eye (Fig. 1B). OCT of the left eye confirmed the presence of macular edema.

Neovascularization and macular edema were treated with intravitreal injections of bevacizumab in both eyes over a two-year follow-up period. During this time, the patient's systemic immunosuppressive therapy, including azathioprine and prednisone, was gradually tapered and discontinued. Follow-up widefield OCTA of the right eye revealed diffuse loss of capillary network architecture involving the macula, along with a regressed neovascular membrane (Fig. 2A). Color-coded *en face* OCTA and a series of cross-sectional OCTA (overlaid on OCT) images (Fig. 2B–F) at the location of RNV lesions were generated using our custom software COOL-ART.[6] In the left eye, macular ischemia and mid-peripheral capillary nonperfusion remained stable. At the final follow-up visit, best-corrected visual acuity was counting fingers in the right eye and 20/25 in the left eye.

2.2. Case 2

An immunocompetent 48-year-old male presented with complaints of floaters in both eyes, with the right eye being worse than the left. He denied systemic symptoms such as fever, oral ulcers, cough, joint pain, or fatigue, and had no past history of systemic diseases. The visual acuity was 20/40 in the right eye and 20/25 in the left eye.

Slit-lamp biomicroscopy revealed mild anterior chamber inflammation in both eyes. Fundus evaluation of the right eye demonstrated sheathed retinal vessels and numerous scattered intraretinal hemorrhages, predominantly in the mid-peripheral inferotemporal region, with a dot-blot appearance (Fig. 3A). The left eye showed fewer intraretinal hemorrhages but exhibited vascular tortuosity and multiple perivascular white, fluffy lesions in the temporal retina (Fig. 3B). FA confirmed periphlebitis with vessel wall leakage, consistent with bilateral occlusive retinal vasculitis. The patient underwent an extensive systemic workup to evaluate potential underlying causes. Serologic testing for viral infections, syphilis, autoimmune diseases, and chest computed tomography were all negative. No evidence of leukemia, sarcoidosis, systemic lupus erythematosus, or other autoimmune conditions was identified.

The condition was initially managed with oral corticosteroids, and methotrexate was introduced as a steroid-sparing agent while prednisone was gradually tapered. After two years of disease stability, the patient developed a recurrence characterized by mild vitritis and focal retinal infiltration in the left eye. This was effectively treated with a periocular injection of triamcinolone acetonide, leading to resolution of the inflammation. Methotrexate was completely discontinued four years later.

Unfortunately, 11 years after the initial presentation, the patient experienced a recurrence of retinal vasculitis involving the temporal macula and nasal periphery in the left eye (Fig. 4). A repeat systemic workup led to a diagnosis of latent TB, confirmed by a strongly positive QuantiFERON-TB Gold test. Chest X-ray and computed tomography showed negative results. Following a complete course of antitubercular therapy and multiple periocular injections of triamcinolone acetonide, the ocular inflammation stabilized. Follow-up widefield OCTA revealed extensive capillary nonperfusion in the right eye, without evidence of neovascular complications (Fig. 5A). In the superior temporal macula of the left eye, capillary nonperfusion, vascular tortuosity, and collateral vessel formation are present in the area (Fig. 5B) of previous active retinal vasculitis on FA (Fig. 4B). At the final follow-up, his visual acuity was 20/25 in the right eye and 20/40 in the left eye.

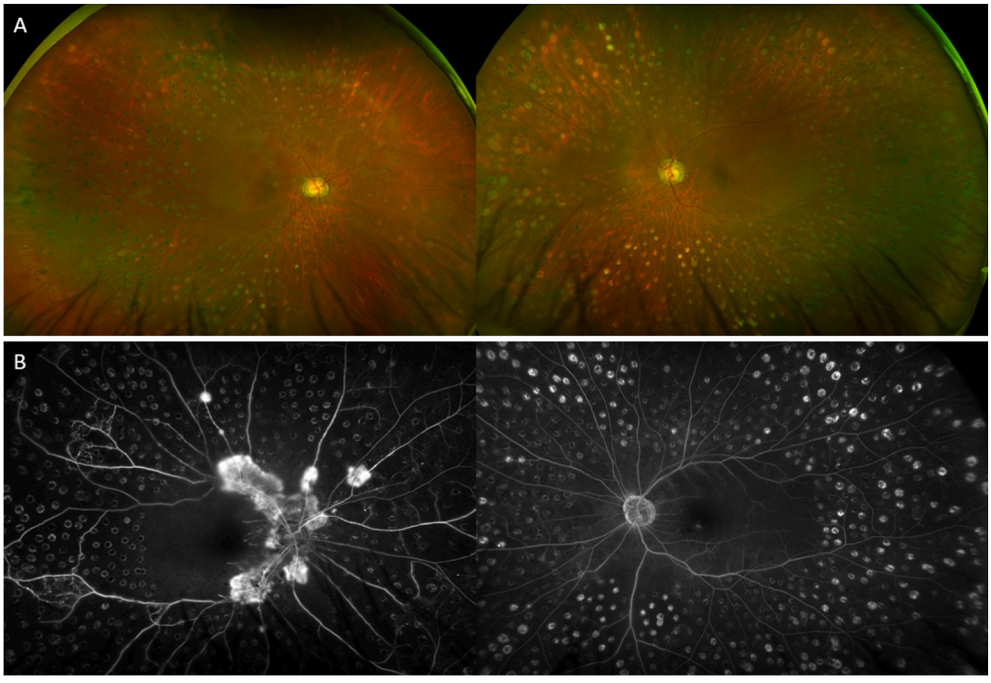


Fig. 1. Ultra-widefield color fundus photography and fluorescein angiography in Case 1, demonstrating features of chronic tubercular retinal vasculitis. (A) Color fundus images of the right eye show retinal hemorrhages, vascular sheathing, ghost vessels, and prominent neovascularization along the vascular arcades. The left eye demonstrates attenuated retinal vessels without evidence of active hemorrhage. (B) Fluorescein angiography of the right eye reveals extensive retinal nonperfusion, multiple areas of segmental vascular leakage, and large retinal neovascularization. In the left eye, macular leakage with parafoveal ischemia and multiple regions of mid-peripheral capillary nonperfusion are observed. Chorioretinal scars from prior panretinal photocoagulation are visible in both eyes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

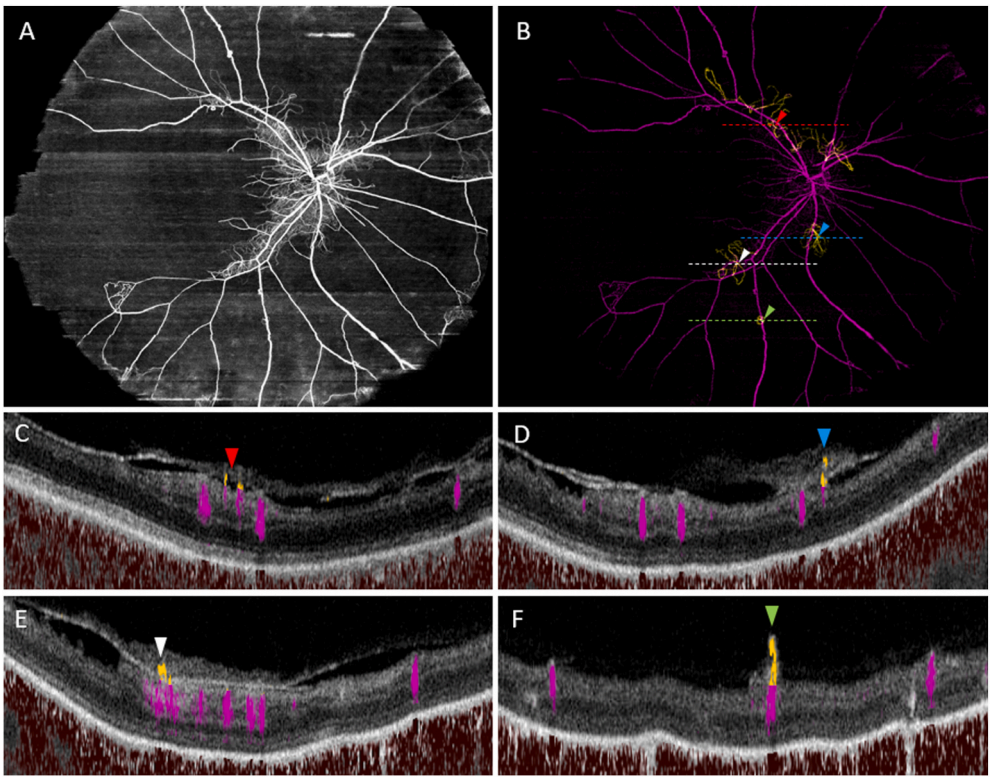


Fig. 2. Sequelae of tubercular retinal vasculitis imaged using widefield OCTA. (A) *En face* OCTA with whole retinal slab of the right eye shows diffuse loss of capillary network architecture. The extent and the distribution of nonperfusion areas, shown as black regions lacking capillaries, are more apparent on the OCTA compared to the FA, where fluorescence from laser scars make the visual assessment of nonperfusion more difficult. (B) Regressed retinal neovascularizations (RNVs; yellow) following multiple anti-VEGF injections along the vascular arcades can also be more clearly visualized on *en face* OCTA due to their location above the inner limiting membrane (ILM) and vascular pathology is not confounded by leakage, in contrast to the retinal vasculature located below (purple). (C–F) Cross-sectional OCT images with overlaid flow signals show RNV profiles with abnormal flow breaching the ILM, corresponding to the red, blue, white, and green arrowheads at the respective scan locations indicated by the dashed lines of matching colors in (B). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

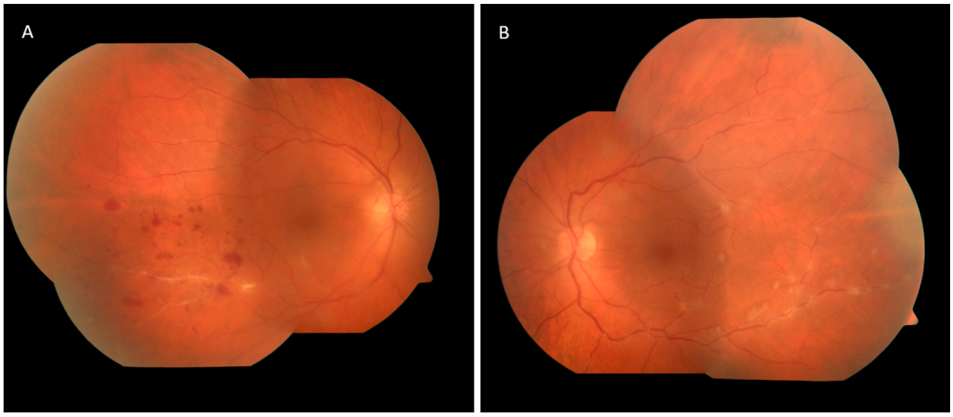


Fig. 3. Montage color fundus photographs in Case 2 demonstrating bilateral occlusive retinal vasculitis with predominant venous involvement and associated vitritis. (A) The right eye shows sheathed vessels and numerous scattered intraretinal hemorrhages in the inferotemporal region. (B) The left eye displays attenuated vessels with segmental periphlebitis and multiple white, fluffy perivascular lesions in the temporal region. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.3. Case 3

A 49-year-old man with no known systemic diseases presented with bilateral blurry vision. His best-corrected visual acuity was 20/25 in the right eye and 20/20 in the left eye. Slit-lamp examination revealed a mild anterior chamber reaction in both eyes.

Fundus examination demonstrated inferior snowballs, ghost vessels, and vascular sheathing in both eyes. Retinal neovascularization was noted in the right eye at the 9 o'clock position, just anterior to the posterior pole. FA revealed multifocal peripheral vascular leakage in both eyes and capillary nonperfusion with neovascularization in the right eye. Indocyanine green angiography confirmed the presence of bilateral choroiditis. Widefield OCTA showed diffuse capillary non-perfusion in the inferotemporal retina, with retinal neovascularization

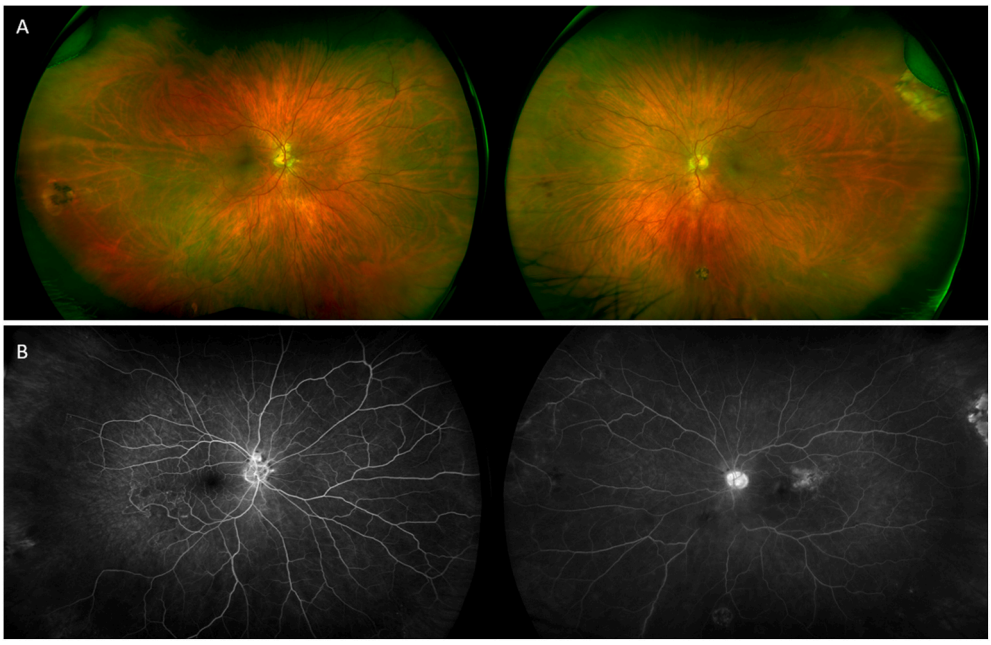


Fig. 4. Recurrence of tubercular retinal vasculitis in the left eye. (A) Ultra-widefield color fundus photograph of the right eye shows inactive retinal vasculitis with ghost vessels. A chorioretinal atrophic patch with pigment clumps, secondary to healed chorioretinitis, is observed in the inferotemporal region. The left eye demonstrates recurrent retinal vasculitis involving the nasal periphery and temporal macula, characterized by sheathed vessels and retinal hemorrhages. (B) Ultra-widefield fluorescein angiography of the right eye reveals temporal capillary dropout and ischemia located beyond the mid-periphery. The left eye shows peripheral phlebitis with vascular leakage in the nasal periphery, focal leakage over the temporal macula, and a nonperfusion area in the inferotemporal far periphery. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

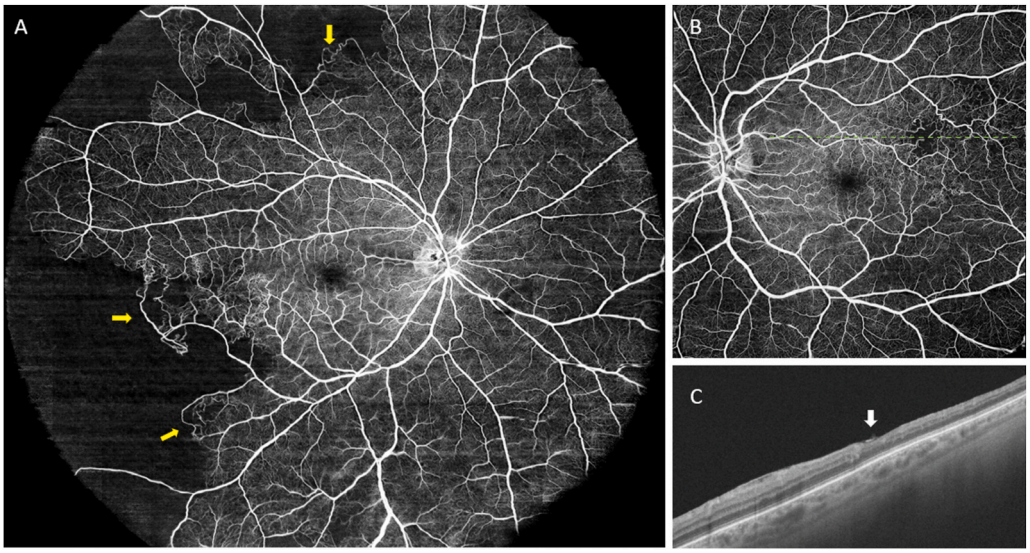


Fig. 5. Resolved and inactive tubercular retinal vasculitis imaged using widefield OCTA, capturing high-resolution images of the retinal vasculature and capillary networks. (A) *En face* OCTA with whole retinal slab of the right eye shows extensive areas of capillary plexus nonperfusion with evidence of vascular remodeling and arteriovenous collaterals formation (yellow arrow). (B) In the left eye, macular capillary disorganization is observed corresponding to previous retinal vasculitis area. Findings include focal nonperfusion with capillary closure and reduced perfusion. (C) The corresponding structural B-scan (dotted line in B) reveals focal inner retinal atrophy and epiretinal membrane formation (white arrow). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

in the right eye corresponding to FA findings (Fig. 6).

A positive QuantiFERON-TB Gold test supported a diagnosis of presumed tubercular retinal vasculitis, while the rest of the systemic workup showed negative results. Chest X-ray and computed tomography showed no evidence of active pulmonary tuberculosis but demonstrated multiple bilateral nodules. The patient was initiated on antitubercular therapy, including rifampin, isoniazid, pyrazinamide, and ethambutol, along with oral prednisone. Following treatment, vascular sheathing improved. At the final follow-up, best-corrected visual acuity was 20/20 in both eyes.

3. Discussion

TRV is characterized by inflammation of the peripheral retinal veins

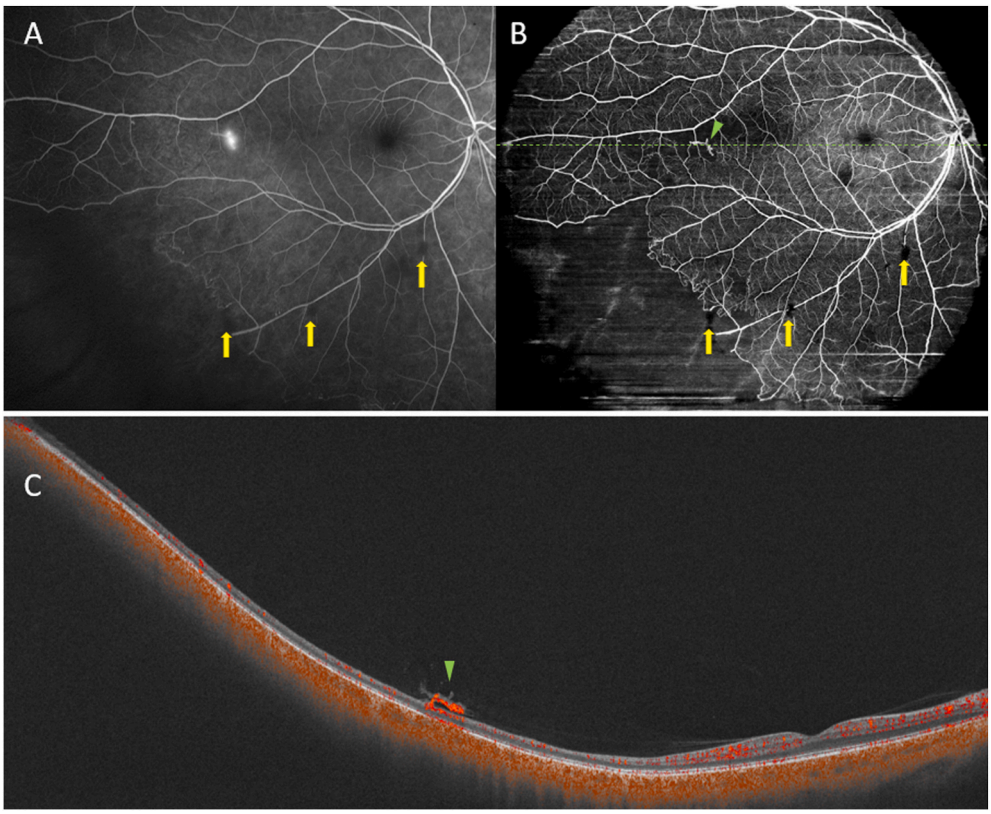


Fig. 6. Widefield OCTA and fluorescein angiography (FA) demonstrating capillary nonperfusion and retinal neovascularization in tubercular retinal vasculitis. (A) FA shows extensive peripheral capillary nonperfusion and a focal area of leakage from neovascularization. (B) *en face* OCTA with combined whole retinal and vitreoretinal interface slab reveals excellent correspondence with FA findings in identifying capillary nonperfusion, while providing superior delineation of perfused versus nonperfused areas. Unlike FA, OCTA is not affected by dye leakage, allowing for clearer visualization of retinal neovascularization (green arrowhead). Note the shadow artifact caused by vitreous opacities (yellow arrow). (C) Corresponding structural OCT B-scan with flow overlay (dotted line in B) shows abnormal flow breaching the internal limiting membrane. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

and associated nonperfusion in the setting of laboratory testing supportive of TB with or without retinal neovascularization, vitreous hemorrhage, and chorioretinal scarring.[7] The key determinant of visual function remains whether macular ischemia is present.[8] Traditionally, fluorescein angiography (FA) has been the gold standard for visualizing retinal vascular leakage, ischemia, and retinal neovascularization. In this series, we have identified presented analogous features using widefield OCTA including capillary nonperfusion, vascular remodeling with arteriovenous collaterals, and abnormal neovascular tufts. These findings highlight the utility of widefield OCTA to demonstrate areas of peripheral nonperfusion areas and retinal neovascularization detected by widefield OCTA are consistent with those observed on FA in TRV.

Widefield OCTA enables visualization of the midperipheral retina and beyond using an eccentric fixation target revealing vascular pathologies that might be missed with standard OCTA imaging of the posterior pole. In this series, the 26 × 21-mm single-shot scan captured all four retinal quadrants and extended to cover approximately 130 degrees of eye angle in one acquisition.[9] The corresponding field of view, based on retinal anatomy, encompasses the posterior pole within the vascular arcades and extends into the mid-peripheral region. Unlike montage techniques, which require combining multiple scans and are limited by patient cooperation, this approach provides a more efficient and reliable widefield view.[10,11]

One of the key advantages of OCTA over FA is its ability to provide volumetric imaging. This is particularly useful for detecting retinal neovascularization, as it allows for the sensitive identification of neovascular lesions by visualizing flow signals that breach the internal limiting membrane.[12–14] In addition, OCTA enables clear visualization of neovascular networks and areas of nonperfusion without the obscuration caused by dye leakage, which is common in FA.[15] FA is limited by its need for intravenous access, long image acquisition time, and the requirement to select one eye as the transit eye. Our findings extend this evidence to TRV, showing that widefield OCTA provides reliable and comparable detection of areas of nonperfusion relative to FA, while offering noninvasive visualization of microvascular remodeling and neovascular morphology. However, OCTA does not replace FA for assessing inflammatory leakage, a key indicator of disease activity that cannot be visualized on OCTA, and this limitation should be acknowledged. Nevertheless, OCTA holds potential as a complementary tool by providing comprehensive structural, flow, and B-scan information that may reveal compensatory changes related to inflammation, particularly when imaging is performed at sites corresponding to leakage seen on angiography.

A limitation of our case series is the overlap in clinical features between TRV and Eales disease. The etiology of Eales disease remains controversial and is considered multifactorial, generally thought to represent an immunologic reaction to an exogenous antigen.[16] Exposure to *Mycobacterium tuberculosis* and hypersensitivity to tuberculoproteins have also been implicated in its pathogenesis. Epidemiologically, Eales disease is rarely reported in the Western world. In Cases 2 and 3, there was no history of exposure to endemic regions, and the relatively young male demographic resembles the typical profile of Eales disease, which adds diagnostic challenging in distinguishing it from TRV. However, both cases presented with clinically evident anterior segment inflammation and vitritis, features more predictive of active TRV, which involves concomitant ocular inflammation and retinal vascular changes. Both patients underwent chest X-ray and computed tomography imaging to evaluate for pulmonary tuberculosis. In Case 2, results were

negative; in Case 3, although no active pulmonary lesions were detected, multiple bilateral nodules were identified, suggesting the possibility of old healed lesions and prior systemic involvement. On laboratory evaluation, Interferon-Gamma Release Assay (IGRA) testing was positive in both cases, with Case 2 showing a strongly positive QuantiFERON-TB Gold result. Regarding ocular imaging, both cases exhibited an occlusive periphlebitis phenotype, and indocyanine green angiography in Case 3 revealed evidence of choroiditis, further supporting TRV activity. Considering OCTA findings, as reported by Nakamura et al., a patient with Eales disease who tested negative for QuantiFERON-TB Gold also demonstrated peripheral nonperfusion and retinal neovascularization consistent between FA and widefield OCTA. This highlights that OCTA findings alone cannot distinguish the two entities. Further OCTA-based studies with larger cohorts are warranted to evaluate whether OCTA-derived biomarkers may provide discriminative value between TRV and Eales disease. However, based on the Collaborative Ocular Tuberculosis Study consensus guidelines,[3] antitubercular therapy was initiated, resulting in clinical improvement and disease stabilization. Given the positive QuantiFERON-TB gold testing and the improvement with antitubercular therapy, we propose the use of a presumed diagnosis of tuberculous etiology in this series.

In conclusion, widefield OCTA detected areas of nonperfusion and neovascularization reliably in TRV. Given that the pathology in TRV often affects the peripheral retina, the widefield imaging capability of OCTA is useful for comprehensive assessment of peripheral microvascular alterations. Its noninvasive and repeatable characteristics make OCTA particularly valuable as a complementary modality to FA for longitudinal disease monitoring.

CRediT authorship contribution statement

An-Lun Wu: Writing – original draft, Investigation. **Yukun Guo:** Visualization, Investigation. **Jinyi Hao:** Visualization, Investigation. **Erick Rivera-Grana:** Writing – review & editing, Data curation. **Steven T. Bailey:** Writing – review & editing. **Yali Jia:** Writing – review & editing, Methodology, Investigation. **Thomas S. Hwang:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization.

Patient consent

This study was approved through the institutional IRB. Written consents were obtained using the forms provided by the IRB.

Funding

This work was supported by grants from National Institutes of Health (R01 EY 036429, R01 EY035410, R01 EY024544, R01 EY027833, R01 EY031394, R43EY036781, P30 EY010572, T32 EY023211, UL1TR002369); the Jennie P. Weeks Endowed Fund; the Malcolm M. Marquis, MD Endowed Fund for Innovation; Unrestricted Departmental Funding Grant and Dr. H. James and Carole Free Catalyst Award from Research to Prevent Blindness (New York, NY); Edward N. & Della L. Thome Memorial Foundation Award, and the Bright Focus Foundation (G2020168, M20230081).

Declaration of competing interest

Yukun Guo: Optovue/Visionix (P), Genentech/Roche (P, R), Ifocus Imaging (P); Steven T. Bailey: Optovue/Visionix (F); Yali Jia: Optovue/Visionix (P, R), Genentech/Roche (P, R, F), Ifocus Imaging (I, P), Optos (P), Boeringer Ingelheim (C), Kugler (R). These potential conflicts of interest have been reviewed and managed by OHSU. Other authors declare no conflicts of interest related to this article.

References

1. Agarwal A, Mahajan S, Khairallah M, Mahendradas P, Gupta A, Gupta V. Multimodal imaging in ocular tuberculosis. *Ocul Immunol Inflamm*. Feb 2017;25(1):134–145. https://doi.org/10.1080/09273948.2016.1231332.
2. Gupta V, Gupta A, Rao NA. Intraocular tuberculosis–an update. *Surv Ophthalmol*. Nov-Dec 2007;52(6):561–587. https://doi.org/10.1016/j.survophthal.2007.08.015.
3. Agrawal R, Testi I, Bodaghi B, et al. Collaborative ocular tuberculosis study consensus guidelines on the management of tubercular uveitis-report 2: guidelines for initiating antitubercular therapy in anterior uveitis, intermediate uveitis, panuveitis, and retinal vasculitis. *Ophthalmology*. Feb 2021;128(2):277–287. https://doi.org/10.1016/j.ophtha.2020.06.052.
4. Jia Y, Bailey ST, Hwang TS, et al. Quantitative optical coherence tomography angiography of vascular abnormalities in the living human eye. *Proc Natl Acad Sci U S A*. May 5 2015;112(18):E2395–E2402. https://doi.org/10.1073/pnas.1500185112.
5. Jia Y, Tan O, Tokayer J, et al. Split-spectrum amplitude-decorrelation angiography with optical coherence tomography. *Opt Express*. Feb 13 2012;20(4):4710–4725. https://doi.org/10.1364/OE.20.004710.
6. Guo Y, Camino A, Zhang M, et al. Automated segmentation of retinal layer boundaries and capillary plexuses in wide-field optical coherence tomographic angiography. *Biomed Opt Express*. Sep 1 2018;9(9):4429–4442. https://doi.org/10.1364/BOE.9.004429.
7. Agrawal R, Gunasekeran DV, Gonzalez-Lopez JJ, et al. Peripheral retinal vasculitis: analysis of 110 consecutive cases and a contemporary reappraisal of tubercular etiology. *Retina*. Jan 2017;37(1):112–117. https://doi.org/10.1097/IAE.0000000000001239.
8. Karampelas M, Sim DA, Chu C, et al. Quantitative analysis of peripheral vasculitis, ischemia, and vascular leakage in uveitis using ultra-widefield fluorescein angiography. *Am J Ophthalmol*. Jun 2015;159(6):1161–1168 e1. https://doi.org/10.1016/j.ajo.2015.02.009.
9. Choudhry N, Duker JS, Freund KB, et al. Classification and guidelines for widefield imaging: recommendations from the international widefield imaging study group. *Ophthalmol Retina*. Oct 2019;3(10):843–849. https://doi.org/10.1016/j.oret.2019.05.007.
10. Hormel TT, Liang GB, Wei X, et al. Visualizing features with wide-field volumetric OCT angiography. *Opt Express*. Mar 11 2024;32(6):10329–10347. https://doi.org/10.1364/OE.510640.
11. Jia Y, Hormel TT, Hwang TS, et al. Widefield OCT angiography. *Prog Retin Eye Res*. Jul 2025;107, 101378. https://doi.org/10.1016/j.preteyeres.2025.101378.
12. Tsuboi K, Mazloumi M, Guo Y, et al. Utility of En Face OCT for the Detection of Clinically Unsuspected Retinal Neovascularization in Patients with Diabetic Retinopathy. *Ophthalmol Retina*. Aug 2023;7(8):683–691. https://doi.org/10.1016/j.oret.2023.03.002.
13. Tsuboi K, Mazloumi M, Guo Y, et al. Early sign of retinal neovascularization evolution in diabetic retinopathy: a longitudinal OCT angiography study. *Ophthalmol Sci*. Mar-Apr 2024;4(2), 100382. https://doi.org/10.1016/j.xops.2023.100382.
14. Wu AL, Guo Y, Hormel TT, et al. Wide-field OCTA quantified peripheral nonperfusion areas predict the risk of subclinical neovascularization. *Eye (Lond)*. Jul 2 2025. https://doi.org/10.1038/s41433-025-03891-2.
15. You QS, Guo Y, Wang J, et al. Detection of clinically unsuspected retinal neovascularization with wide-field optical coherence tomography angiography. *Retina*. May 2020;40(5):891–897. https://doi.org/10.1097/IAE.0000000000002487.
16. Das T, Pathengay A, Hussain N, Biswas J. Eales' disease: diagnosis and management. *Eye (Lond)*. Mar 2010;24(3):472–482. https://doi.org/10.1038/eye.2009.315.